

QUBO Formulation: Mars Satellite Observation Scheduling Problem

Mars Satellite Observation Scheduling Problem

Problem Statement. Given a set of N candidate observation windows for a Mars-orbiting instrument, each candidate $i \in \{0, \dots, N-1\}$ is associated with a science priority score $p_i \geq 0$ and a resource consumption value $r_i \geq 0$. A set of conflict pairs $\mathcal{C} \subseteq \{0, \dots, N-1\} \times \{0, \dots, N-1\}$ identifies observations that cannot be executed simultaneously due to temporal overlap or incompatible spacecraft/instrument resources. The total available resource capacity is denoted by $R_{\max} > 0$. The scheduling task is to select a feasible subset of observations that maximizes the aggregate science priority score while satisfying all pairwise conflict constraints and the global resource capacity limit.

Decision Variables. For each candidate observation $i \in \{0, \dots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if observation i is scheduled, and 0 otherwise. The decision vector is denoted $\mathbf{x} = (x_0, x_1, \dots, x_{N-1})^\top \in \{0, 1\}^N$.

QUBO Derivation. Step 1 – Original Objective. The scheduling problem seeks to maximize the total priority score:

$$\max_{\mathbf{x} \in \{0,1\}^N} \sum_{i=0}^{N-1} p_i x_i. \quad (1)$$

Since QUBO solvers minimize energy, we negate the objective to obtain the minimization form:

$$\min_{\mathbf{x} \in \{0,1\}^N} \left(- \sum_{i=0}^{N-1} p_i x_i \right). \quad (2)$$

Step 2 – Constraints. The problem is subject to two types of constraints. First, for every conflicting pair $(i, j) \in \mathcal{C}$, at most one observation may be selected:

$$x_i + x_j \leq 1, \quad \forall (i, j) \in \mathcal{C}. \quad (3)$$

Second, the total resource consumption must not exceed the available capacity:

$$\sum_{i=0}^{N-1} r_i x_i \leq R_{\max}. \quad (4)$$

Step 3 – Penalty Conversion. We convert the inequality constraints into quadratic penalty terms added to the objective. Let $\lambda > 0$ be the penalty weight.

- *Conflict constraints:* The condition $x_i + x_j \leq 1$ is violated only when $x_i = x_j = 1$. This is enforced by the penalty term $\lambda x_i x_j$. Summing over all conflicts yields:

$$P_{\text{conflict}}(\mathbf{x}) = \lambda \sum_{(i,j) \in \mathcal{C}} x_i x_j. \quad (5)$$

- *Resource constraint:* We penalize deviations from the capacity limit using a squared penalty:

$$P_{\text{resource}}(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} r_i x_i - R_{\max} \right)^2. \quad (6)$$

Expanding the square and applying the idempotent property $x_k^2 = x_k$ for binary variables gives:

$$\begin{aligned} P_{\text{resource}}(\mathbf{x}) &= \lambda \left(\sum_{i=0}^{N-1} r_i^2 x_i + \sum_{i \neq j} r_i r_j x_i x_j - 2R_{\max} \sum_{i=0}^{N-1} r_i x_i + R_{\max}^2 \right) \\ &= \sum_{i=0}^{N-1} \lambda (r_i^2 - 2R_{\max} r_i) x_i + \lambda \sum_{i \neq j} r_i r_j x_i x_j + \lambda R_{\max}^2. \end{aligned} \quad (7)$$

Step 4 – Combined Hamiltonian. Adding the negated objective and both penalty terms yields the QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = - \sum_{i=0}^{N-1} p_i x_i + \lambda \sum_{(i,j) \in \mathcal{C}} x_i x_j + \sum_{i=0}^{N-1} \lambda (r_i^2 - 2R_{\max} r_i) x_i + \lambda \sum_{i \neq j} r_i r_j x_i x_j + \lambda R_{\max}^2. \quad (8)$$

Step 5 – Q Matrix Entries and Offset. The Hamiltonian is expressed in the standard QUBO form $H(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c$. By collecting coefficients for each monomial $x_i x_j$ (with $i \leq j$), we obtain the general closed-form expressions for the upper-triangular Q matrix entries and the constant offset:

$$Q_{i,i} = -p_i + \lambda (r_i^2 - 2R_{\max} r_i), \quad \forall i \in \{0, \dots, N-1\}, \quad (9)$$

$$Q_{i,j} = \lambda \cdot \mathbb{I}((i,j) \in \mathcal{C}) + 2\lambda r_i r_j, \quad \forall 0 \leq i < j \leq N-1, \quad (10)$$

$$c = \lambda R_{\max}^2, \quad (11)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation increases the energy more than the maximum possible gain from the objective function. The maximum possible increase in the original objective from scheduling any single observation is $\max_i p_i$. A single violated conflict or resource constraint incurs a penalty of at least λ . Therefore, selecting $\lambda > \max_i p_i$ guarantees that the energy of any infeasible solution strictly exceeds that of the best feasible solution. In practice, λ is set to a value safely above the sum of all priority scores or a known upper bound on $\max_i p_i$ to provide a robust margin against numerical precision limits.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0, 1\}^N$ recovers the optimal schedule because $H(\mathbf{x})$ is constructed as $H(\mathbf{x}) = -\sum_{i=0}^{N-1} p_i x_i + P(\mathbf{x})$, where $P(\mathbf{x}) \geq 0$ is zero if and only if $\mathbf{x}$ satisfies all conflict and resource constraints. For any feasible $\mathbf{x}$, $H(\mathbf{x})$ equals the negative objective value. For any infeasible $\mathbf{x}$, $P(\mathbf{x}) \geq \lambda$. Since λ is chosen to exceed the maximum possible objective gain, it follows that $H(\mathbf{x}_{\text{infeasible}}) > H(\mathbf{x}_{\text{feasible}})$ for all feasible $\mathbf{x}_{\text{feasible}}$. Consequently, the global minimum of H must occur at a feasible point that minimizes the negative objective, which is equivalent to maximizing the original science priority score.

Illustrative Example. Consider a concrete instance with $N = 3$ candidate observations. Let the priority scores be $p_0 = 10$, $p_1 = 20$, and $p_2 = 15$. The resource usages are $r_0 = 2$, $r_1 = 3$, and $r_2 = 4$. The maximum resource capacity is $R_{\max} = 5$. All pairs conflict, so $\mathcal{C} = \{(0, 1), (0, 2), (1, 2)\}$. We set the penalty weight to $\lambda = 1000$.

Substituting these values into the general formulas from Step 5 yields the concrete Q matrix entries:

$$Q_{0,0} = -10 + 1000(2^2 - 2 \cdot 5 \cdot 2) = -16010, \quad (12)$$

$$Q_{1,1} = -20 + 1000(3^2 - 2 \cdot 5 \cdot 3) = -21020, \quad (13)$$

$$Q_{2,2} = -15 + 1000(4^2 - 2 \cdot 5 \cdot 4) = -24015, \quad (14)$$

$$Q_{0,1} = 1000 + 2 \cdot 1000 \cdot 2 \cdot 3 = 13000, \quad (15)$$

$$Q_{0,2} = 1000 + 2 \cdot 1000 \cdot 2 \cdot 4 = 17000, \quad (16)$$

$$Q_{1,2} = 1000 + 2 \cdot 1000 \cdot 3 \cdot 4 = 25000, \quad (17)$$

$$c = 1000 \cdot 5^2 = 25000. \quad (18)$$

The resulting Hamiltonian is:

$$H(x_0, x_1, x_2) = -16010x_0 - 21020x_1 - 24015x_2 + 13000x_0x_1 + 17000x_0x_2 + 25000x_1x_2 + 25000. \quad (19)$$

Evaluating H over the feasible space (where at most one x_i can be 1 due to conflicts, and

resource sum ≤ 5):

$$H(0, 0, 0) = 25000, \tag{20}$$

$$H(1, 0, 0) = -16010 + 25000 = 8990, \tag{21}$$

$$H(0, 1, 0) = -21020 + 25000 = 3980, \tag{22}$$

$$H(0, 0, 1) = -24015 + 25000 = 985. \tag{23}$$

The global minimum occurs at $\mathbf{x}^* = (0, 0, 1)$ with energy $H(\mathbf{x}^*) = 985$. This corresponds to selecting observation 2, which yields the maximum feasible science priority score of 15.